Chapter 9

Basic Concepts of Probability

Learning Outcomes

When you will complete this chapter, you would be able to-

- ❖Basic terms and content of probability
- ❖Calculate probability between two events
- ❖Estimate and interpret different types of rules of probability
- ❖Identify the relation and different types of probability
- ❖Identify the relation and different types of probability with practical examples

Lecture Contents

- Basic Terminology of Probability
- Different Approaches of Probability
- Application and Significance of Probability
- Laws of Probability
- Probability Rules
- Joint, Marginal and Conditional Probability
- Various Mathematical Problems of Probability

Basic Terminology of Probability

Probability: A value between zero and one, inclusive, describing the relative possibility (chance or likelihood) an event will occur.

Experiment: Experiment is an act that can be repeated under given conditions.

Trial: Unit of an experiment is known as trial. This means that trial is a special case of experiment. Experiment may be a trial or two or more trials.

Outcomes: The result of an experiment is known as outcomes.
Example: Throwing a die is a trial and getting 1 or 2 or 3 or 4 or 5 or 6 is an outcome.

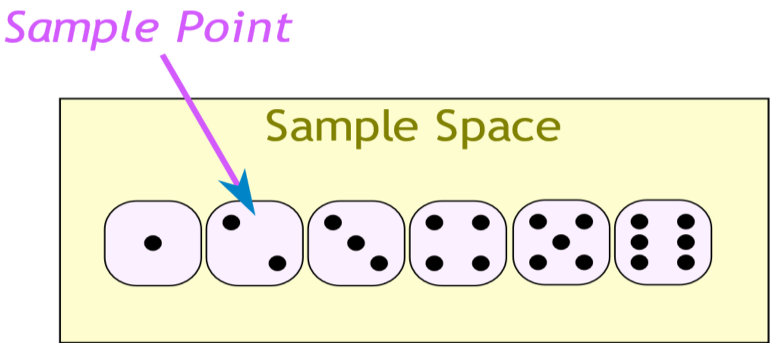
Equally likely Outcomes: Outcomes of a trial are said to be equally likely if we have no reason to expect any one rather than the other. Example-1) In tossing a fair coin, the outcomes head and tail are equally likely, 2) In throwing a balanced die all the six faces are equally likely.

Basic Terminology of Probability

Mutually Exclusive Outcomes: Outcomes or cases are said to be mutually exclusive if the happening of any one of them precludes the happening of all others. Example-1) In tossing a coin, the outcomes hear and tail are mutually exclusive.2) In throwing a die, the six outcomes which are the different points on the faces of the die is mutually exclusive.

Exhaustive outcomes: Outcomes of an experiment are said to be exhaustive if they include all possible outcomes. Example-in throwing a die exhaustive number of outcomes are 6.

Sample space: The collection of all possible outcomes of a random experiment is called sample space. Sample space is usually denoted by S or Ω . Example:1) If we toss a coin, the sample space is, Ω ={H,T}. Where H and T denote the hear and tail of the coin, respectively, 2)If a six–sided die is thrown, the sample space is, Ω ={1,2,3,4,5,6}.



Probability

Probability:

If there are n mutually exclusive, equally likely and exhaustive outcomes of an experiment and if m of these outcomes are favorable to an event A, then the probability of the event A which is denoted by $P(A)$ is defined by

$P(A)$ = Favorable outcomes of an event A/Total number of outcomes of the experiments

$$P(A) = \frac{m}{n}$$

Example: After tossing a coin what is the probability that we will get head?

Solution:

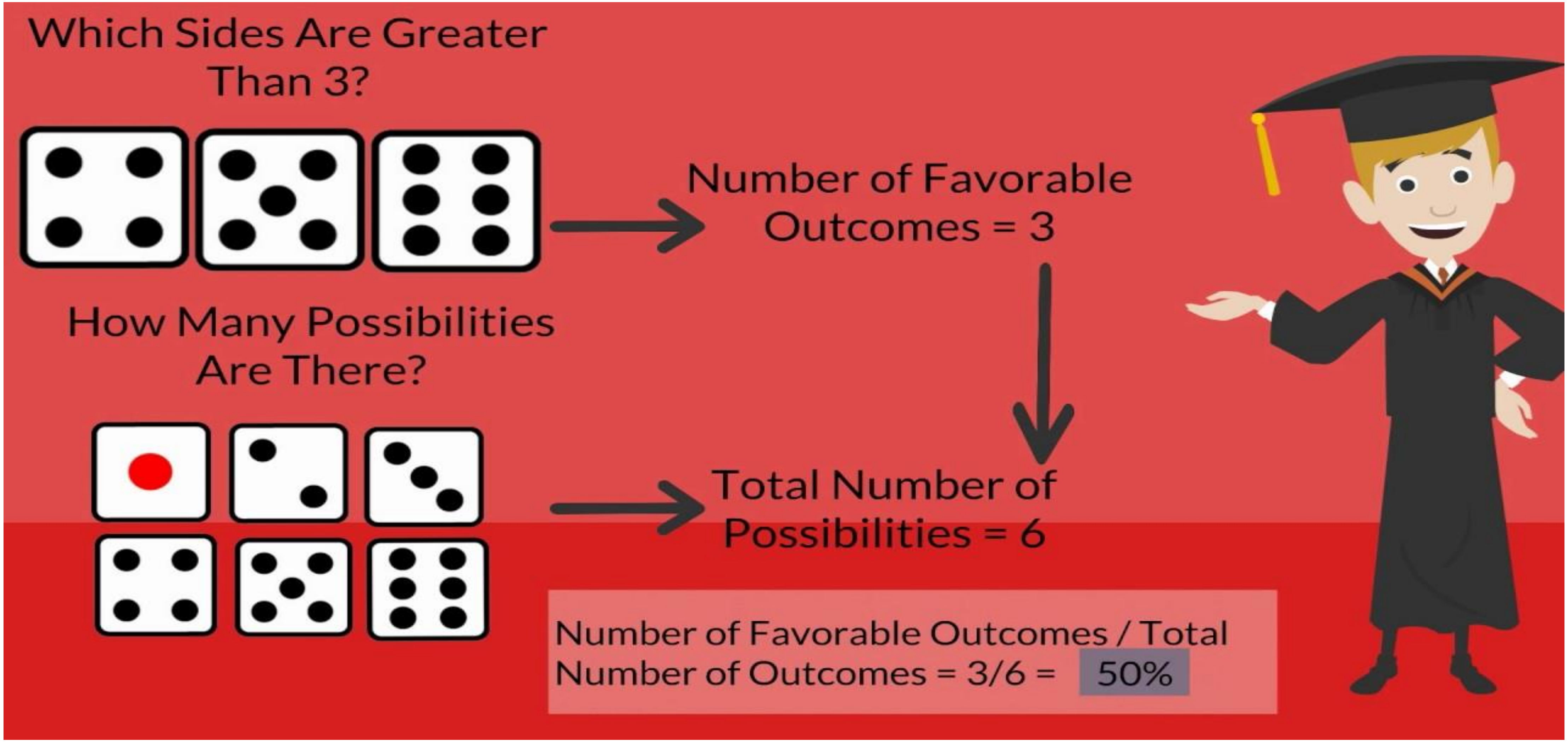
We know,

$$P(A) = \frac{m}{n}$$

Here total number of outcome is n= 2 and favorable outcome, m=1

So $P(A) = \frac{m}{n} = \frac{1}{2} = 0.5$

Probability



Probability

Example: A bag contains 4 white and 6 black balls. If one ball is drawn at random from the bag, what is the probability that it is a) black, b) white c) white or black d) red?

Solution:

Total number of balls is 10.

a) Let A be the event that the ball is black, then the number of outcomes favorable to A is 6. Hence,

$$P[A]= \frac{No.of\ Black\ Balls}{Total\ No.of\ Balls} = \frac{6}{10} = 0.6$$

b) Let B be the event that the ball is white, then the favorable outcomes corresponding to B are 4. Therefore,

$$P[A]= \frac{No.of\ White\ Balls}{Total\ No.of\ Balls} = \frac{4}{10} = 0.4$$

c) Let C be the event that the ball is white or black, then the favorable outcomes corresponding to C are 10. Therefore,

$$P[A]= \frac{No.of\ White\ or\ Black\ Balls}{Total\ No.of\ Balls} = \frac{10}{10} = 1$$

d) Let D be the event that the ball is red, then the favorable outcomes corresponding to D are 0. Therefore,

$$P[A]= \frac{No.of\ Red\ Balls}{Total\ No.of\ Balls} = \frac{0}{10} = 0$$

Application and Significance of Probability

Probability is the backbone of statistics. It has vast application in every field of statistics. The various practical applications of the probability are:

a)In statistical inference, the problem of estimation and the test of hypothesis are based on probability theory.
b)The various test of significance, viz- Z-test, Chi-square test, F- test, t- test are derived from the theory of probability.
c) Decision theories are based on fundamental laws of probability and expected value.
d) The use of probability theory is increasing in economic decision making.
e) The use of subjective probabilities is made when actual measurement is not possible.

Laws of Probability

There are two laws which are very important.

1. **All probabilities are between 0 and 1 inclusive**

i.e. $0\leq P(A) \geq 1$

2. **The sum of all the probabilities in the sample space is 1**

i.e. $\sum_{i=1}^{n} P(A)= 1$

There are some other laws which are also important.

□ The probability of an event which cannot occur is 0.

The probability of any event which is not in the sample space is zero.

□ The probability of an event which must occur is 1.

The probability of the sample space is 1.

□ The probability of an event not occurring is one minus the probability of it occurring.

Addition Rules

Additive Law of Probability: There are two rules of addition, the special rule of addition and the general rule of addition.

Special Rule of Addition: To apply the special rule of addition, the events must be mutually exclusive. If two events A and B are mutually exclusive, the special rule of addition states that the probability of one or the other event's occurring equals the sum of their probabilities. This rule is expressed in the following formula:

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$$

General Rule of Addition: When two events both occur, the probability is called a joint probability. In this situation, we use the general rule of addition. If A and B are two events in , then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Addition Rules

Example: Mr. Ali feels that the probability that he will pass mathematics is 2/3 and statistics is 5/6.If the probability that he will pass both the course is 3/5, what is the probability that he will pass at least one of the course?

Solution: Let M and S be the events that he will pass the course mathematics and statistics, respectively. The event M U S means that at least one of M or S occurs. Therefore, according to the addition rule we get,

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= 2/3 + 5/6 - 3/5$$
$$= 9/10$$

Addition Rules

Example:
In a sample of 500 students, 320 said they had a stereo, 175 said they had a TV, and 100 said they had both. 5 said they had neither. If a student is selected at random, what is the probability that the student has only a stereo or TV? What is the probability that the student has both a stereo and TV?

Solution: Let S and T be the events that students had stereo and TV, respectively. Then the probability that student has only stereo or TV is-

$$\begin{aligned} P\,(S \text{ or } T) &= P(S) + P\,(T) - P\,(S \text{ and } T) \\ &= 320/500 + 175/500 - 100/500 \\ &= 0.79 \end{aligned}$$

The probability that the student has both a stereo and TV:

$$\begin{aligned} P\,(S \text{ and } TV) &= 100/500 \\ &= 0.20 \end{aligned}$$

Joint Probability

Joint probability is the probability of two events in conjunction. That is, it is the probability of both events together. The joint probability of A and B is written P ($A \cap B$), P(AB) or P(A,B).

Example: The question, "Do you like watching TV?" was asked of 100 people. Results are shown in the table. What is the probability of a randomly selected individual being a male who likes watching TV?

	Yes	No	Total
Male	19	41	60
Female	12	28	40
Total	31	69	100

Solution: This is just a joint probability. The number of "Male and like watching TV" divided by the total = 19/100 = 0.19

Marginal Probability

Marginal probability is the probability of A, regardless of whether event B did or did not occur. If B can be thought of as the event of a random variable X having a given outcome, the marginal probability of A can be obtained by summing the joint probabilities over all outcomes for X.

Example: The question, "Do you like watching TV?" was asked of 100 people. Results are shown in the table. What is the probability of a randomly selected individual like watching TV?

	Yes	No	Total
Male	19	41	60
Female	12	28	40
Total	31	69	100

Solution: Since no mention is made of gender, this is a marginal probability, the total who like watching TV divided by the total = 31/100 = 0.31

Conditional Probability

The probability of an event occurring given that another event has already occurred is called a conditional probability. The probability of event B occurring that event A has already occurred is read "the probability of B given A" and is written: P (B|A)

If A and B are two events in a probability space, such that P[A]>0,then the conditional probability of B given A, denoted by P(B|A)is defined by-

$$P(B|A) = P(A \text{ and } B) / P(A) = \frac{P(A \cap B)}{P(A)}, \text{ if } P(A)>0$$

Similarly, conditional probability of A given B is defined by

$$P(A|B) = P(A \text{ and } B) / P(B) = \frac{P(A \cap B)}{P(B)}, \text{ if } P(B)>0$$

Conditional Probability

Example: The question, "Do you like watching TV?" was asked of 100 people. Results are shown in the table. What is the probability of a randomly selected individual is a male if it is given that he likes watching TV?

	Yes	No	Total
Male	19	41	60
Female	12	28	40
Total	31	69	100

Solution: The conditional probability M given Y is-

$$P(M|Y) = \frac{P(M \cap Y)}{P(Y)} = \frac{\frac{19}{100}}{\frac{31}{100}} = \frac{19}{100} \times \frac{100}{31} = \frac{19}{31} = 0.6$$

Example

A hamburger chain found that 75% of all customers use salad, 80% use ketchup and 65% use both. What are the probabilities that

i. a ketchup user uses salad
ii. a salad user uses ketchup

Solution

Let A be the event "customer uses salad" and B be the event "customer uses ketchup". Thus, we have, $p(A)=0.75$, $p(B)=0.80$ and $p(A\cap B)=0.65$.

The probability that a ketchup user uses salad is the conditional probability of event A, given event B is

$$p(A|B)=\frac{p(A\cap B)}{p(B)}=\frac{0.65}{0.80}=0.8125.$$

So, the probability that a ketchup user uses salad

The probability that a salad user uses ketchup is the conditional probability of event B, given event A is

$$P(B/A)=\frac{P(A\cap B)}{P(B)}=\frac{0.65}{0.75}=0.87$$

Similarly, a salad user uses ketchup 0.87

Conditional Probability

Example: A bag contains red and blue marbles. Two marbles are drawn without replacement. The probability of selecting a red marble and then a blue marble is 0.28. The probability of selecting a red marble on the first draw is 0.5. What is the probability of selecting a blue marble on the second draw, given that the first marble drawn was red?

Solution:

$$\text{P (Blue | Red)} = \frac{P\ (Blue\ and\ Red)}{P\ (Red)} = \frac{0.28}{0.5} = 0.56$$

Multiplication Rule

Specific Multiplication Rule:

The Specific Rule of Multiplication requires that two events A and B are *independent*.

This rule is written: $P\ (A \text{ and } B) = P\ (A)*P(B)$

$$P(A \cap B) = P(A) * P(B)$$

General Multiplication Rule

The General Rule of Multiplication requires that two events A and B are *dependent*

P (A and B) = P (A) * P (B|A)

$P(A \cap B) = \text{P (A) * P(B|A)}$

Multiplication Rule

Example:

1. If the probability that person A will be alive in 0.7 and the probability that person B will be alive in 0.5, what is the probability that they will both be alive in 20 years?

These are independent events, so

$P(E_1 \text{ and } E_2) = P(E_1) \times P(E_2) = 0.7 \times 0.5 = 0.35$

[Note, however, that if person A knows person B, then they will be **dependent** events, especially if A is married to B.]

Example

There are 10 rolls of film in a box, 3 of which are defective. Two rolls are to be selected one after another. What is the probability of selecting a defective roll followed by another defective roll?

Solution

The first roll of film selected from the box being found defective is event D_1. $\therefore p(D_1)=\frac{3}{10}$.

The second roll selected being found defective is event D_2. Therefore, $p(D_2|D_1)=\frac{2}{9}$. Since, after the first selection was found to be defective, only 2 defective rolls of film remained in the box containing 9 rolls.

So the probability of two defectives is

$$= p(D_1 \text{ and } D_2)$$

$$= p(D_1) * p(D_2|D_1)$$

=3/10*2/9=.066

Complement Rule

The complement rule is used to determine the probability of an event occurring by subtracting the probability of the event not occurring from 1 i., e. $p(A) = 1 - p(\bar{A})$.

Example

Weight	Event	Probability
Underweight	A	0.025
Satisfactory	B	??
Overweight	C	0.075

Find $p(B)$.

Solution

We know that $p(B) = 1 - p(\bar{B}) = 1 - (0.025 + 0.075) = 0.90$.

Chapter Exercise

1. The following table gives a two-way classification of all basketball players at a state university who began their college careers between 2001 and 2005, based on gender and whether or not they graduated.

	Graduated	Did Not Graduate	Total
Male	126	55	181
Female	133	32	165
Total	259	87	346

If one of these players is selected at random, find the following probabilities.

a. P (female or did not graduate)
b. P (graduated and male)
c. P (Graduated | female)

Chapter Exercise

1.In a class of 120 students, 60 are studying English, 50 are studying French and 20 are studying both English and French. If a student is selected at random from this class, what is the probability that he is studying English if it is given that he is studying French?

2. Jack can hit a target 5 out of 7 chances. Andy can hit a target 6 out of 11.Find the chances that the target is hit once they both try?

3. Russell is playing in a cricket match and a game of football at the weekend. The probability that his team will win the cricket match is 0.7, and the probability of winning is 0.9 in the football. What is the probability that his team will win in both matches? What is the probability that his team will not win in both matches?

4. Suppose you have a box with 3 blue marbles, 2 red marbles, and 4 yellow marbles. You are going to pull out one marble, record its color, put it back in the box and draw another marble. Find the probability that the first marble is red and second marble is yellow.

5. The probability that Mr. X will die in the next 20 years is 1/5 and the probability that Mr. Y will die in the next 20 years is 1/7. What is the probability that

i. Both will die in the next 20 years. ii. At least one will die in the next 20 years. iii. Neither will die in the next 20 years.

Thank You